NETWORKING PRACTICAL ASSIGNMENT

Ping, Traceroute and Local Network Discovery

Objective: Test communication between devices on the same Wi-Fi, trace the route to a website, observe a disconnected device, and identify local IP addresses.

Note: This is a sample student-style report. Example IPs and outputs are illustrative; replace them with your own results and screenshots before submitting.

Task 1 — Test communication using ping

First, connect the laptop and a friend's phone to the same Wi-Fi. Check the phone's local IP in Wi-Fi details, then run the command below on Windows, replacing the example IP with the phone's actual address.

```
ping 192.168.1.25
```

How to read it: A reply means the target responded. 'Request timed out' means no reply arrived in time; check Wi-Fi, the IP address and firewall settings. 'Destination host unreachable' means the sender or network could not reach that address. Some phones block ping even while connected.

Result	Meaning
Reply from 192.168.1.25	The target responded to ping.
Request timed out	No response before timeout; check settings.
Destination host unreachable	No usable path to the target address.

Observation: Record the actual replies and packet statistics from your own device; do not copy sample output as a real result.

Task 2 — Run traceroute to YouTube

Traceroute lists network hops between your device and a destination. On Windows, use `tracert`; on macOS/Linux, use `traceroute`.

```
tracert www.youtube.com
```

macOS/Linux:

```
traceroute www.youtube.com
```

Each numbered line is a hop and may show response times. Some routers do not answer these probes, so asterisks do not always mean the internet is broken.

Hop	Example	Explanation
1	192.168.1.1	Usually the home router/default gateway
2	ISP router	A router inside the internet provider's network
3 onward	More routers or * * *	Intermediate hops; some may not respond

Actual responding hop count: _____ (fill this from your own output).

Task 3 — Disconnect a device and ping again

First, ping the phone while both devices are connected to Wi-Fi. Then disconnect the phone from Wi-Fi and run the same ping command again.

Expected result: The old phone IP will normally stop replying. The exact error may vary by Windows and router settings. If the phone reconnects later, it may receive a different IP address.

Condition	Likely result	Meaning
Both devices on Wi-Fi	Ping replies, if allowed	The phone is reachable locally
Phone Wi-Fi disconnected	Request timed out	The old IP is no longer reachable

Important: Client isolation or a firewall can also block ping, so a failed ping alone does not prove the network is faulty.

Task 4 — List devices using arp -a

Open Command Prompt on Windows and type:

```
arp -a
```

This shows IP and MAC addresses learned recently by your computer. It may not list every connected device. Compare the results with your router's connected-device list or each device's Wi-Fi information.

IP address (example)	MAC address (example)	Device / owner
192.168.1.1	AA-BB-CC-11-22-01	Home router
192.168.1.25	AA-BB-CC-11-22-25	Friend's phone — verify
192.168.1.30	AA-BB-CC-11-22-30	My laptop — verify

All addresses in this table are examples. Use the entries from your own arp -a output.

Task 5 — One-page report: network testing

For this practical, I connected my laptop and another device to the same Wi-Fi network. I checked the second device's IP address and used the ping command in Command Prompt to test whether it replied. If replies arrive, the target is reachable at the IP level. If the command times out, I would confirm both devices are on the correct Wi-Fi, check the IP address, and consider firewall or client-isolation settings.

Next, I used `tracert www.youtube.com` on Windows (or `traceroute www.youtube.com` on macOS/Linux) to see the route towards YouTube. The output lists hops and response times. Some routers may not respond, so asterisks do not always indicate a fault. I would record the hop count from my own output because routes can change by network and time.

I then disconnected one device from Wi-Fi and repeated the ping test. The device would normally stop replying because it was no longer reachable on the local network. Finally, I ran `arp -a` to view IP and MAC addresses learned by the laptop. I compared these entries with the router's connected-device list to identify each gadget. The ARP list may not include every device.

Evidence to attach: Screenshots of ping, tracert, ping after disconnecting a device, and arp -a. Write down the actual IP addresses and hop count from your own network.

Conclusion: These commands help test local connectivity, understand the route to a website, and view nearby IP-to-MAC mappings. Results can vary with router settings, firewalls and the internet provider.